Task

Interview Task: AI-Based Shopping Assistant for Drone Parts

1. Background & Objective

Drone builders in India rely on a handful of specialty electronics retailers (Robu.in, ThinkRobotics, Zbotic, Evelta, etc.) for parts like flight controllers, companion computer HATs, and RF modules. Stock availability is unpredictable — a part can be "out of stock" for weeks and reappear with no announcement, or go up for **pre-order** with a vague ETA.

Your task is to design and build a lightweight AI-powered assistant that:

1. **Monitors** a set of retailer websites for product availability changes.
2. **Alerts** users (via Telegram/WhatsApp) when a tracked product becomes available or opens for pre-order.
3. **Lets users chat** with the system to search, filter, and compare products, and to add new products/URLs to the watchlist — all through the same messaging interface.

This is a systems-design exercise as much as an AI exercise. You are expected to make and justify real engineering trade-offs, not just wire together an LLM and a scraper.

2. Scope of the System

2.1 Target sources (seed list — extensible)

- robu.in
- **thinkrobotics.com**
- zbotic.in
- evelta.com
- (Design the source layer so more sites can be added without a rewrite.)

2.2 Seed watchlist (must be tracked from day 1)

- Raspberry Pi 5 (and Pi 5-compatible HATs)
- Holybro/Hex **Pixhawk 6X**
- General category: flight controllers, companion computers, HATs/carrier boards (select any 2)

2.3 Core capabilities

Capability	Description
Daily monitoring	Scheduled job checks each tracked product's page/listing daily (interval should be configurable)
Change detection	Detect: in-stock ↔ out-of-stock transitions, new pre-order listings, price changes (optional/bonus), new matching products appearing in category/search pages
Lightweight LLM classification	Use a small/cheap LLM (not a large frontier model) to parse noisy/unstructured page content into structured status (in-stock / out-of-stock / pre-order / unknown) and extract product attributes
Alerting	Push notifications to Telegram and/or WhatsApp when a watched product's status changes favorably (back in stock, pre-order opened)
Conversational chat interface	Same messaging channel lets a user: search products, filter by spec/price/availability, ask about a specific product's status/history, and add new products or URLs to monitor
Dynamic watchlist	Users can add/remove tracked products conversationally — no code change or redeploy required

3. What You Need to Design

You will be evaluated primarily on **how** you designed this system, so make deliberate, explainable choices in these areas:

- Data acquisition layer
- LLM usage — where and why
- State & change detection
- Messaging / chat interface

4. Deliverables

1. **GitHub repository** containing:
   - Full source code, organized and readable
   - `README.md` covering: architecture overview, setup/run instructions, design decisions and trade-offs, known limitations, and how to add a new site/product
   - A simple architecture diagram (can be an image or ASCII/mermaid diagram in the README)
2. **Short video demo** (screen recording is fine) showing:
   - The daily monitoring job running (or a simulated/forced run)
   - A stock/pre-order alert arriving in Telegram or WhatsApp
   - Live chat interaction: searching/filtering products, and adding a new product to the watchlist conversationally

7. Time Expectation

This is intended as a take-home exercise of a few days of focused effort, not a production system. Scope your work accordingly — a well-reasoned, well-documented partial system is preferred over a rushed complete one.